Locking mechanism

```c
#include "kernel/types.h"
#include "kernel/stat.h"
#include "user/user.h"

int
main(void)
{
  int lk = lock_create();
  if(lk < 0){
    printf("locktest: lock_create failed\n");
    exit(1);
  }

  printf("Lock test starting (lock id = %d)\n", lk);

  int pid = fork();
  if(pid < 0){
    printf("locktest: fork failed\n");
    lock_destroy(lk);
    exit(1);
  }

  // ---- both parent and child reach this point ----

  int me = getpid();
  for(int i = 0; i < 3; i++){
    if(lock_acquire(lk) < 0){
      printf("Process %d: lock_acquire failed\n", me);
      exit(1);
    }

    // --- critical section ---
```

```c
    // --- critical section ---
    printf("Process %d: >>> INSIDE critical section (iteration %d)\n", me, i);
    // small busy loop to make the overlap more visible in output
    for(volatile int j = 0; j < 100000; j++)
      ;
    printf("Process %d: <<< LEAVING critical section (iteration %d)\n", me, i);
    // --- end critical section ---

    if(lock_release(lk) < 0){
      printf("Process %d: lock_release failed\n", me);
      exit(1);
    }
  }

  if(pid == 0){
    // child exits
    exit(0);
  } else {
    // parent waits for child
    int status;
    wait(&status);

    if(lock_destroy(lk) < 0){
      printf("locktest: lock_destroy failed\n");
      exit(1);
    }
    printf("Lock test done. Lock %d destroyed.\n", lk);
  }

  exit(0);
}
```

```
sravan@PRASADSTOREA:/mnt/c/Users/prasa/OneDrive/Desktop/OS_folder$ cd OS_Projects_G5/
sravan@PRASADSTOREA:/mnt/c/Users/prasa/OneDrive/Desktop/OS_folder/OS_Projects_G5$ cd Project_1/
sravan@PRASADSTOREA:/mnt/c/Users/prasa/OneDrive/Desktop/OS_folder/OS_Projects_G5/Project_1$ cd xv6-riscv/
sravan@PRASADSTOREA:/mnt/c/Users/prasa/OneDrive/Desktop/OS_folder/OS_Projects_G5/Project_1/xv6-riscv$ ls
LICENSE  Makefile  README  fs.img  kernel  mkfs  sysproc.c  test-xv6.py  user
sravan@PRASADSTOREA:/mnt/c/Users/prasa/OneDrive/Desktop/OS_folder/OS_Projects_G5/Project_1/xv6-riscv$ make clean
rm -f *.tex *.dvi *.idx *.aux *.log *.ind *.ilg \
*/*.o */*.d */*.asm */*.sym \
kernel/kernel fs.img \
mkfs/mkfs .gdbinit \
        user/usys.S \
user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh
user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_testppid user/_t
estps user/_testtrace user/_testwaitx user/_testcount user/_dorphan user/_pwd user/_locktest user/_threadtest user/_t
estprocinfo user/_shmtest
sravan@PRASADSTOREA:/mnt/c/Users/prasa/OneDrive/Desktop/OS_folder/OS_Projects_G5/Project_1/xv6-riscv$ make qemu
riscv64-linux-gnu-gcc -march=rv64gc -g -c -o kernel/entry.o kernel/entry.S
```

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ locktest
Lock test starting (lock id = 0)
Process 3: >>> INSIDE critical section (iteration 0)
Process 3: <<< LEAVING critical section (iteration 0)
Process 3: >>> INSIDE critical section (iteration 1)
Process 3: <<< LEAVING critical section (iteration 1)
Process 3: >>> INSIDE critical section (iteration 2)
Process 3: <<< LEAVING critical section (iteration 2)
Process 4: >>> INSIDE critical section (iteration 0)
Process 4: <<< LEAVING critical section (iteration 0)
Process 4: >>> INSIDE critical section (iteration 1)
Process 4: <<< LEAVING critical section (iteration 1)
Process 4: >>> INSIDE critical section (iteration 2)
Process 4: <<< LEAVING critical section (iteration 2)
Lock test done. Lock 0 destroyed.
$
```